

Impact of Housing Conditions and Air Quality Awareness with Indoor PM_{2.5} Levels in Urban Jodhpur

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Abstract

Indoor air pollution poses a major public health concern in India, particularly among urban populations exposed to high levels of particulate matter. This study aimed to assess the influence of housing conditions and awareness on indoor particulate matter (PM_{2.5}) levels in Jodhpur, Rajasthan. A cross-sectional study was conducted across 12 households representing slum, low-, middle-, and high-income groups, as well as commercial and industrial areas. Indoor PM_{2.5} concentrations were continuously monitored over a one-year period using pre-calibrated AirVisual Pro devices. Awareness regarding indoor air pollution and preventive measures was evaluated through structured questionnaires administered before and after an awareness intervention. The findings revealed that effective ventilation and good hygiene practices significantly contributed to reducing indoor PM_{2.5} levels. Poorly ventilated homes, such as House 1 (62 µg/m³) and House 4 (73 µg/m³), exhibited higher pollutant concentrations, while well-ventilated or air-conditioned homes, such as House 2 (46 µg/m³) and House 3 (40 µg/m³), maintained lower levels. The highest concentration (111 µg/m³) was recorded in House 5, which is located near a highway with inadequate ventilation. The awareness component showed a marked improvement in residents' understanding of air pollution sources, health impacts, and mitigation strategies following real-time monitoring and education sessions. The study concludes that housing ventilation, seasonal behavioral factors, and real-time monitoring play a crucial role in determining indoor air quality. Strengthening building design standards and enhancing public awareness are vital for sustainable indoor air quality management in urban India.

Keywords: Housing conditions, indoor air quality (IAQ), PM_{2.5}, seasonal variations

INTRODUCTION

In India, exposure to ambient air pollution on the urban population is among the highest levels anywhere else in the world. Indoor air pollution is a critical public health concern, especially in developing countries. It is caused by harmful chemicals and materials that degrade air quality within enclosed spaces, making it up to 10 times worse than outdoor air pollution.^[1] The urban landscape, comprising densely packed houses that have been built without any consideration for proper ventilation, combined with the use of biomass as cooking fuel and high atmospheric dust, contaminates the indoor air to very hazardous levels. These pollutants contribute to severe health outcomes, such as respiratory infections, chronic obstructive pulmonary disease (COPD), lung cancer, and adverse perinatal outcomes. Incriminating evidence is gradually being generated about the role of pollutants in type-2 diabetes mellitus, attention deficit hyperactivity disorder, and cognitive function disorders.^[2]

Women and children are disproportionately affected due to their extended exposure to indoor pollutants, particularly from cooking with biomass fuels. These findings underscore the urgent need for interventions such as clean cooking technologies, improved building designs, and public education to mitigate the health impacts of indoor air pollution.^[1-3]

A study conducted in South India assessed the levels of PM_{2.5} in rural households, finding significant differences based on the type of housing and ventilation available. The study revealed that houses with poor ventilation and those using biomass

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fuels had higher concentrations of PM2.5, posing greater health risks to residents.^[3] In future, a significant proportion of households will use incense sticks, mosquito coils, and kerosene, which contribute to poor indoor air quality (IAQ). Additionally, overcrowding and inadequate ventilation were prevalent. Hence, it highlights the importance of improving housing conditions, increasing awareness about the health risks of indoor air pollution, and promoting the use of cleaner fuels to reduce the consequences may give more references to indoor air pollution.^[4]

In India, cultural housing conditions, climate, and geographies vary from region to region. Hence, studies in specific geographies are needed to understand the nature and extent of indoor air pollution. Therefore, this study was conducted to determine the effect of housing conditions and people's awareness about pollution on indoor air pollution in Jodhpur, western Rajasthan, India.

SUBJECTS AND METHODS

Study area: This community-based cross-sectional study was conducted in Jodhpur, a city in western Rajasthan located at 26°17'N and 73°01'E, spread over 233 km², having a population of 9,21,476 (source: Jodhpur, India Population (2024) - Population Stat).^[5] The study was conducted in residential premises, in slums, low-income, middle-income, and upper-income households, as well as commercial and industrial establishments.

Data collection procedure

Indoor air quality

Detailed methodology is provided elsewhere.^[5] For IAQ assessment, two houses each were randomly selected from

slums, low-income, middle-income, and upper-income households, as well as commercial and industrial establishments. Portable electricity/battery-operated pre-calibrated air quality monitors, AirVisual Pro Indoor Sensor (IQAir), were deployed in the selected households for one year. These houses were equipped with air quality monitors to measure PM2.5 levels.

Awareness assessment

A survey was carried out using a structured, pre-tested questionnaire to assess the level of awareness about air pollution among the residents of all households selected for air quality monitoring (AQM), and devices were installed.

The housing conditions in these 12 houses were analyzed using a self-structured questionnaire and observation checklist, where higher scores indicated poorer air quality and lower scores suggested better air quality in the house.

This scoring system offers valuable insights into the current state of IAQ to identify specific areas for intervention and improvement. Scores of 0 typically indicate better practices or conditions that contribute to healthier indoor environments, while scores of 1 highlight areas where corrective actions are needed to mitigate potential health risks associated with indoor air pollution [Supplementary Table 1].

Data management and analysis

All collected information was recorded digitally using an Excel sheet for further analysis.

Ethics and informed consent

Ethical clearance was obtained from the Institute Ethics Committee. Written consent was acquired from each participant after providing a comprehensive explanation of

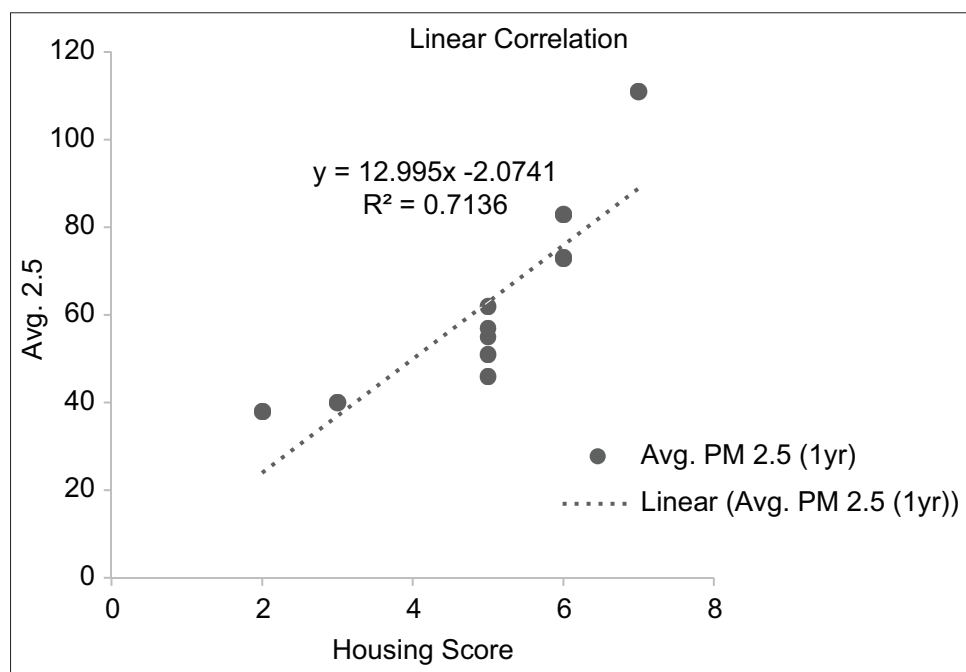


Figure 1: Linear correlation depicting housing characteristics with PM2.5

Table 1: Awareness of participants in selected areas of Jodhpur

Questions		Pre-awareness yes (n=12)	Post-awareness yes (n=12)
Do you know about air pollution?		12 (100.00)	12 (100.00)
Is air pollution harmful to the health of humans and other living beings?		12 (100.00)	12 (100.00)
Do you know which air pollutants cause air pollution?		6 (50.00)	11 (91.67)
Do you know about the health effects of air pollution?		11 (91.67)	11 (91.67)
Which of the following is/are the health effects caused by air pollution?	Breathing problem	7 (58.33)	12 (100.00)
	Cough up	0 (00.00)	11 (91.67)
	Watery eyes	1 (8.33)	11 (91.67)
	Headache	2 (16.67)	11 (91.67)
	All of the above	4 (33.33)	12 (100.00)
Does air pollution cause disease?		11 (91.67)	12 (100.00)
Which of the following is/are the risk factor (s) caused by air pollution?	Respiratory infection	10 (83.33)	12 (100.00)
	Heart disease	0 (00.00)	12 (100.00)
	COPD	1 (8.33)	12 (100.00)
	Lung cancer	0 (00.00)	12 (100.00)
	All of the above	2 (16.67)	12 (100.00)
Do you know about the sources of air pollution?		10 (83.33)	11 (91.67)
What, according to you, is the main source of air pollution?	Industrial chimney	6 (50.00)	11 (91.67)
	Traffic exit	8 (66.7)	9 (75.00)
	Open burning of garbage	1 (8.33)	10 (83.33)
	Building houses	1 (8.33)	9 (75.00)
	Cigarette smoke	3 (25.00)	9 (75.00)
	Other factors	0 (00.00)	11 (91.67)
Do you know the number of deaths in India due to air pollution?		1 (8.33)	9 (75.00)
Do you know which city of Rajasthan is highly air-polluted?		4 (33.33)	1 (8.33)
Do you know about the air quality index?		2 (16.67)	12 (100.00)
Have you noticed air pollution in the area where you live?		9 (75.00)	9 (75.00)
How do you rate the overall air quality in your area?	Good	8 (66.7)	7 (58.33)
	Bad	3 (25.00)	3 (25.00)
	Don't know	0 (00.00)	2 (16.67)
How severe is air pollution in the area where you live?	Very less	1 (8.33)	0 (00.00)
	Less	4 (33.33)	3 (25.00)
	Moderate	5 (41.67)	7 (58.33)
	More	2 (16.67)	2 (16.67)
	The most	0 (00.00)	0 (00.00)
Do you think indoor air pollution can happen?		9 (75.00)	12 (100.00)
Where do you think pollution is more?	Indoor	1 (8.33)	0 (00.00)
	Outdoor	11 (91.67)	12 (100.00)
How do you protect yourself when there is pollution in the air?	I wear a face mask	5 (41.67)	5 (41.67)
	Cover my face with a handkerchief	3 (25.00)	7 (58.33)
	I do nothing	4 (33.33)	0 (00.00)
Do you agree that every citizen's responsibility is to improve the environment?		12 (100.00)	12 (100.00)
What can you do to improve the environment	Planting trees	12 (100.00)	12 (100.00)
	Keeping surroundings clean	0 (00.00)	12 (100.00)
	Planting trees and keeping the surroundings clean	0 (00.00)	12 (100.00)
	Others	0 (00.00)	12 (100.00)
Post-sensor installed	Have you benefited from having an air quality device installed in your home?	–	5 (41.67)
	If the air quality device is installed in your home for a long time, will you take care of it?	–	7 (58.33)
	Do you check the air quality instrument data regularly?	–	7 (58.33)
	Have you tried to work out pollution by looking at data from air quality instruments?	–	7 (58.33)
	If yes above question, which of the following have been attempted (Cleaning)	–	7 (58.33)
	Has your air quality device increased your awareness?	–	12 (100.00)

Table 2: Household characteristics with PM2.5 indoor air quality measurement

Area		LIG		Industrial area		Commercial area		MIG		HIG	
Houses		House 1	House 2	House 3	House 4	House 5	House 6	House 9	House 10	House 11	House 12
Cross ventilation		1	1	0	1	1	1	1	0	1	1
Are there ventilators in living rooms		1	1	0	1	1	0	1	0	1	0
Are the following gadgets used at home?	Air conditioners	1	0	0	1	1	1	0	0	1	0
	Desert coolers	0	1	1	0	0	0	0	0	0	0
	Exhaust fan	1	1	1	1	1	1	1	1	1	1
	Room heater	1	1	1	0	0	1	0	0	1	0
	Blower	0	0	0	0	0	0	0	0	0	0
	Air purifier	0	0	0	0	0	0	0	0	0	0
Thick layer of dust		0	0	0	0	0	0	0	0	0	0
Are dustbins covered with a lid		0	0	0	0	0	0	0	1	0	0
Are there mold/fungi in the household		0	0	0	0	0	0	0	0	0	0
Pakka house construction		0	0	0	0	0	0	0	0	0	0
Is the house located on a highway/expressway		0	0	0	0	1	0	0	0	0	0
Is the house located in an industrial area/commercial area		0	0	0	1	1	1	0	0	0	0
Are there any polluting establishments in the neighborhood (within 50 meters)? (Restaurant/automobile workshop/waste disposal site/other specify)		0	0	0	1	1	1	1	1	0	0
Is kitchen separate from living room		0	0	0	0	0	0	1	1	0	0
Source of ventilation in kitchen		0	0	0	0	0	0	0	1	0	0
Avg. PM 2.5 (1 yr)		62	46	40	73	111	83	51	55	57	38
Score		5	5	3	6	7	6	5	5	5	2

Note: House 7 and House 8 were not analyzed due to their poor structural conditions in slum areas, which made it difficult to determine a proper correlation with PM2.5 levels and other variables. HIG: High-income group, LIG: Low-income group, MIG: Middle-income group

the study's purpose and procedures. Participant confidentiality was strictly maintained throughout the study.

RESULTS

Table 1 shows air pollution awareness among 12 participants in Jodhpur. At baseline, while 100% recognized air pollution and its health risks, specific knowledge was limited: Only 16.67% knew about the air quality index (AQI), and 8.33% were aware of pollution-related deaths in India. After the intervention, awareness improved significantly. AQI knowledge rose to 100%, and 75% knew about mortality due to air pollution. Recognition of specific health effects, such as breathing problems, reached 100%. All participants reported that AQM devices enhanced their understanding, with 58.33% committed to long-term use and 41.67% reporting direct benefits from the devices.

Table 2 shows that indoor PM2.5 levels are closely linked to housing conditions. Poor ventilation and proximity to pollution sources led to higher PM2.5, as seen in House 5 (111 $\mu\text{g}/\text{m}^3$, score 8) in a commercial area. In contrast, better-ventilated homes with lower housing condition scores had reduced PM2.5, such as House 3 (40 $\mu\text{g}/\text{m}^3$, score 4) in an industrial area and House 12 (38 $\mu\text{g}/\text{m}^3$, score 3) in a commercial area with good air conditioning and cleanliness. House 1 in a low-income area recorded 62 $\mu\text{g}/\text{m}^3$ (score 6) due to inadequate

ventilation. Overall, lower scores of better housing conditions correlated with lower PM2.5, and seasonal increases were observed during winter.

Figure 1 presents a strong positive linear correlation. This suggests that as the score for housing characteristics increases, the average PM2.5 levels also tend to increase. The *P* value for this correlation is 0.002, which is statistically significant [Supplementary Table 2].

PM2.5 levels vary by area: the industrial area fluctuated 52.36–110.72 $\mu\text{g}/\text{m}^3$, slum areas had consistently high peaks up to 177.68 $\mu\text{g}/\text{m}^3$, high-income group (HIG) colonies had lower levels, max 79.02 $\mu\text{g}/\text{m}^3$, and middle-income group (MIG) colonies showed moderate levels up to 91.64 $\mu\text{g}/\text{m}^3$. All areas experienced higher PM2.5 in winter [Supplementary Figure 1].

DISCUSSION

This study examines the relationship between housing conditions, awareness, and indoor PM2.5 levels in urban Jodhpur, building on existing research on IAQ. These findings extend the work of Huda *et al.*,^[5] who mapped ambient air quality in Jodhpur using low-cost sensors. While their study focused on outdoor pollution, the present research demonstrates that indoor air is significantly affected by both external emissions and internal housing deficiencies, highlighting the need for integrated indoor-outdoor AQM.

A positive correlation was found between housing condition scores, indicating structural and hygienic inadequacies and higher PM2.5 levels. This aligns with Vijay *et al.*^[6] who reported spatial and temporal variations in indoor PM2.5 across Indian urban homes, attributing higher concentrations to poor ventilation and household density. Similarly, Anand and Phuleria^[7] found elevated indoor PM2.5 in Delhi's slums due to structural constraints.

Proximity to traffic significantly impacted IAQ, as seen in a house near a highway with the highest. This supports Alhajari *et al.*^[8] who identified traffic-related emissions as a major indoor pollutant in Kuwaiti homes. In contrast, a well-maintained commercial-area home with air conditioning recorded low PM2.5, indicating that mechanical ventilation and cleanliness can reduce outdoor infiltration. This contrasts with Ścibor *et al.*^[9] whose study in Poland found high indoor PM2.5 despite moderate outdoor levels, due to aging buildings and lack of filtration—highlighting that both building quality and behavioral practices are critical in urban Indian contexts.

Seasonal variation was evident, with PM2.5 rising in winter due to reduced ventilation and increased heating. This is consistent with Vijay *et al.*^[6] but this study adds a behavioral insight: residents restricted airflow to retain warmth, worsening indoor pollution. This thermal comfort–air quality trade-off is also noted by Niculita-Hirzel *et al.*^[10] in studies on urban environments and respiratory health.

Although cooking with biomass was not reported, findings from Karaca *et al.*^[11] show that even clean fuels like Liquefied Petroleum Gas (LPG) can generate PM2.5 spikes if kitchens lack proper ventilation. This underscores the need for kitchen-specific interventions such as exhaust fans. Papagiannis *et al.*^[12] identified traffic, construction, and industrial emissions as major PM2.5 sources in Dushanbe, Tajikistan, a pattern mirrored in Jodhpur's industrial and commercial zones, especially when combined with poor housing infrastructure.

Awareness improved dramatically after installing air quality monitors. Awareness of the AQI rose from 16.67% to 100%, supporting Hou *et al.*^[13] who found that real-time data improves health literacy in Taiwan. Al-Shidi *et al.*^[14] reported similar gains in Oman through structured campaigns, but this study shows that device feedback alone can drive awareness, suggesting a low-effort, high-impact educational strategy. However, baseline knowledge of pollution-related mortality was very low, a gap also found by Gabel *et al.*^[15] in Greenland, indicating a global shortfall in communicating pollution's health burden.

A clear socioeconomic gradient was observed higher-income households having better ventilation and lower PM2.5. This reflects findings by Dwivedi *et al.*^[16] who noted disproportionate exposure among women in low-income homes, and Ronita Bardhan *et al.*^[17] who linked urban re-densification in Mumbai to deteriorated housing conditions. This study reaffirms that structural inequities place the greatest

pollution burden on slum and low-income residents, who lack resources to improve their living environments.

Strengths and Limitations

The study used real-time, pre-calibrated air quality monitors (AirVisual Pro) over a 12-month period, ensuring reliable longitudinal data. The inclusion of diverse housing types—slum, low-, middle-, and high-income, as well as commercial and industrial areas—enhances the representativeness of findings for urban Jodhpur.

However, the small sample size limited statistical power and generalizability. Findings are specific to Jodhpur and may not apply to other climatic or urban settings. Only PM2.5 was measured, excluding other health-relevant pollutants such as NO₂, CO, and VOCs.

Recommendations

Public education campaigns should expand to include information on the AQI, pollution-related mortality, and health risks, using schools, health centers, and local media. Urban planning policies must enforce ventilation standards in housing, particularly in low-income settlements, to reduce indoor pollution exposure.

CONCLUSION

This study highlights that housing conditions and awareness significantly influence indoor PM2.5 levels in Jodhpur. Poor ventilation and proximity to pollution sources were linked to higher PM2.5, while better ventilation and air conditioning reduced levels. Seasonal spikes, especially in winter, worsened IAQ. Although general awareness of air pollution was high, knowledge of specific indicators like the AQI and pollution-related mortality was low. Installing air quality monitors significantly improved awareness and encouraged long-term engagement in air quality management.

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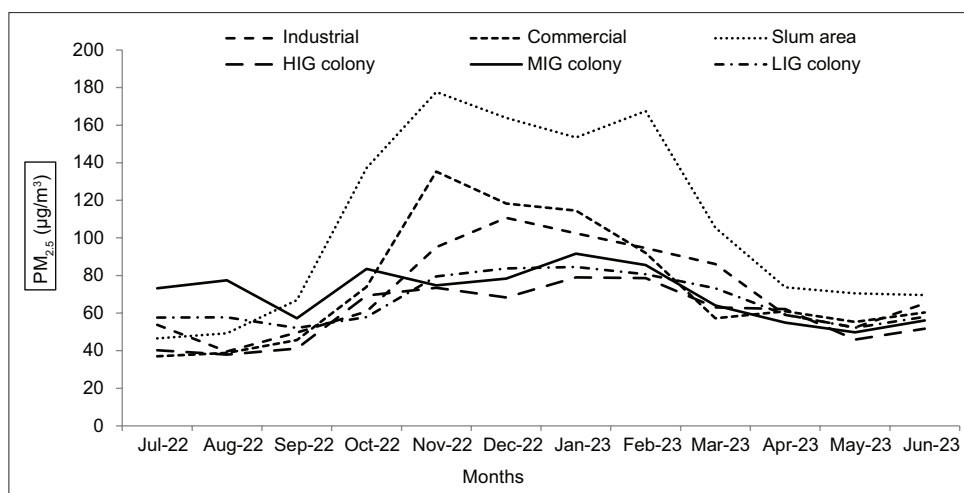
Conflicts of interest

There are no conflicts of interest.

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Supplementary Figure 1: Temporal distribution of PM2.5 levels in indoor areas of all colonies

Supplementary Table 1: Scoring criteria for indoor air quality measurement

S. No.,	Parameters	Criteria	Scoring
1.	Ventilation	Cross ventilation present	Yes=0, No=1
		Ventilators in living rooms	Yes=0, No=1
		Ventilators in the kitchen	Yes=0, No=1
2.	Appliance use	Air conditioners	Yes=0, No=1
		Desert coolers	Yes=0, No=1
		Exhaust fans	Yes=0, No=1
		Air purifier	Yes=0, No=1
3.	Heating appliances	Room heater	Yes=1, No=0
		Blower	Yes=1, No=0
4.	Dust and cleanliness	The presence of thick dust	Yes=1, No=0
		Mold/fungi presence	Yes=1, No=0
5.	Housing condition	Pakka house construction	Yes=0, No=1
		The kitchen is separated from the living room	Yes=0, No=1
		Dustbins covered	Yes=0, No=1
6.	Building and location factors	Proximity to the polluting establishment	Yes=1, No=0
		Located on a highway/expressway	Yes=1, No=0
		Located in an industrial/commercial area	Yes=1, No=0

Supplementary Table 2: Correlation of housing characteristics with PM2.5

Avg. PM2.5 (1 yr)	Score	Person correlation	P
62.00	5.00	0.845	0.002
46.00	5.00		
40.00	3.00		
73.00	6.00		
111.00	7.00		
83.00	6.00		
51.00	5.00		
55.00	5.00		
57.00	5.00		
38.00	2.00		